Machine Learning Topics: Imbalanced Datasets

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Topics

1. Motivating Examples
2. Techniques
3. Sampling
4. Cost-sensitive Algorithms
5. One Class Learning
6. Evaluation

What are Imbalanced Datasets ?

Some Applications

Most security challenges:

- Fraud detection, e.g., recall the discussion of spear phishing in our unsupervised learning (DM) module
- Malware detection
- Intrusion detection

Detection of rare diseases
Subsetting operations, selection

Available in Python toolbox Imbalanced-learn

Techniques for Imbalanced Datasets

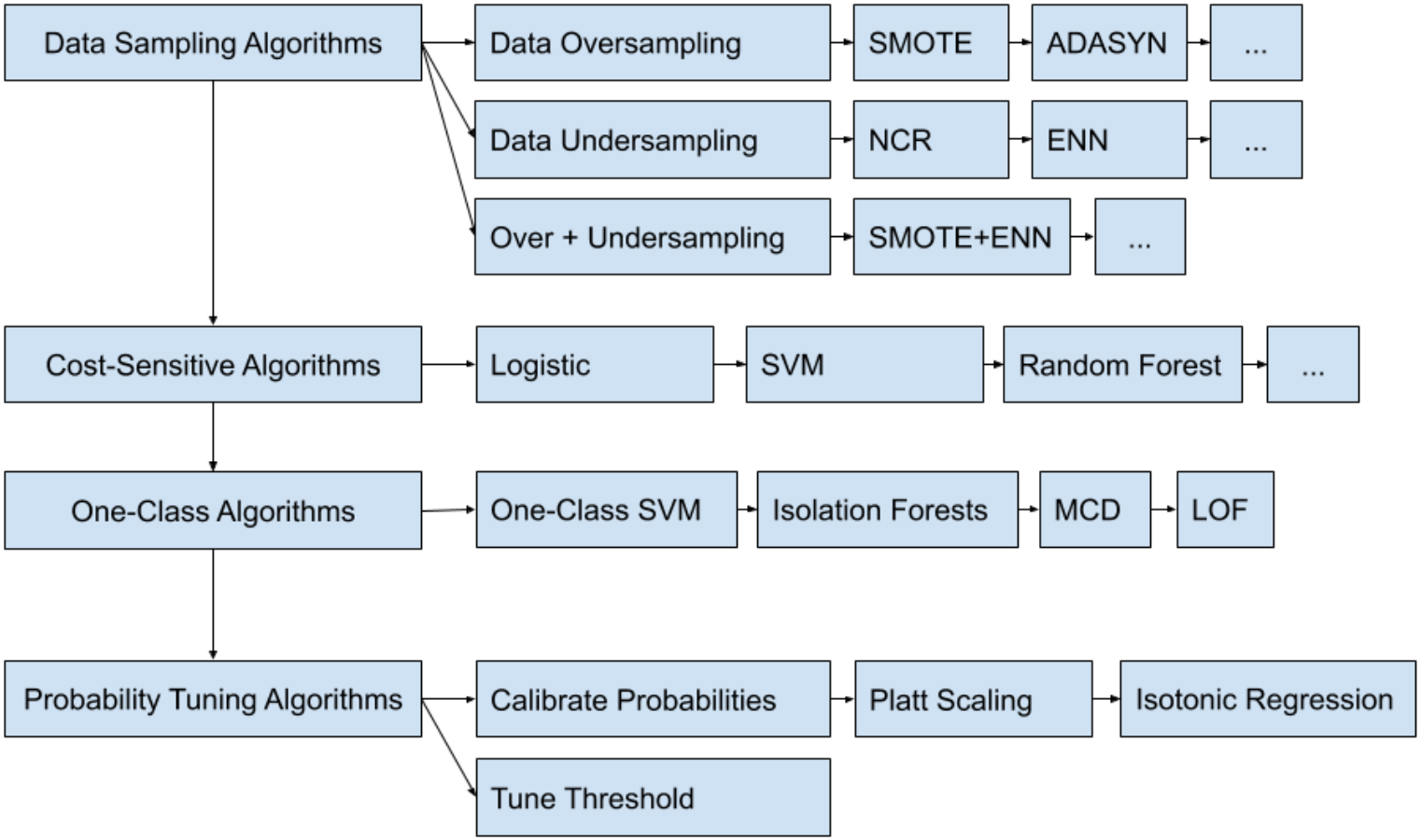
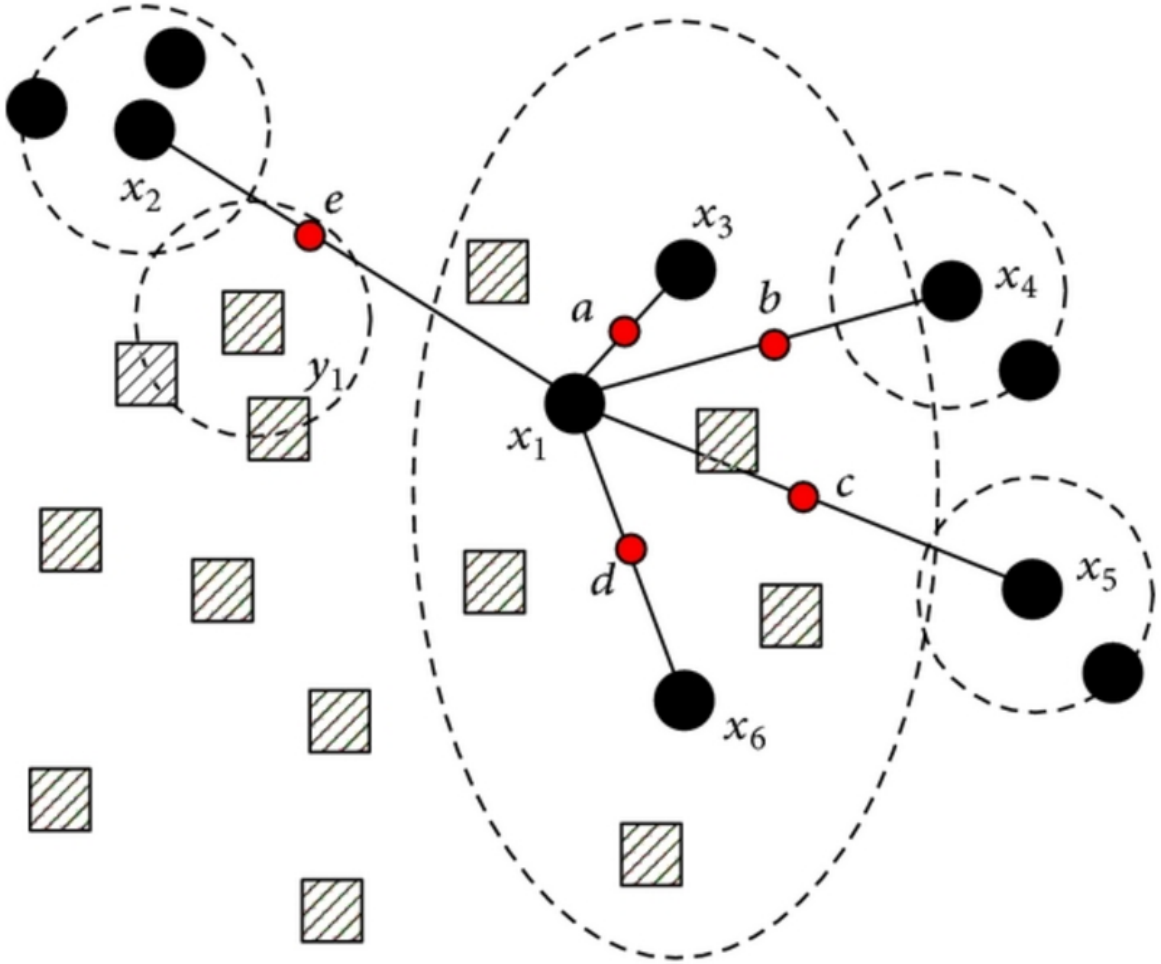


Figure courtesy of Jason Brownlee

Data Sampling

1. Oversampling of minority class: SMOTE, ADASYN
2. Undersampling of majority class: NCR, ENN
3. Hybrid methods: over + undersampling, e.g., SMOTE+ENN
4. Ensemble methods

SMOTE: Synthetic Minority Oversampling Technique



Cost Sensitive Versions

- SVM: Boundary Movement, Biased Penalties, and extend hinge loss function
- Logistic
- Random Forest

A General Technique

Metacost

Learn From a Single Class

- One class SVM: hypersphere, separate the origin by a hyperplane
- Isolation forests
- Minimum Covariance Determinant
- Local Outlier Factor

Metrics for Imbalanced Datasets

Accuracy is not a good metric for imbalanced datasets! Why?

- Geometric Mean
- Matthews Correlation Coefficient
- Cost-sensitive evaluation
- Balanced Detection Rate and its normalized version[2]

[2] http://ceur-ws.org/Vol-2124/